

Magnetic Pressure Erases the Lyman Edge

Sub-Eddington 3D MHD disk simulation shows suppressed Lyman edge, matching observed smooth AGN continua.

arXiv:2609.22421 — Geometrically thin sub-Eddington AGN accretion disks show suppressed Lyman edge.

arXiv:2609.22421 doi:10.48550/arXiv.2609.22421



Background

Classical optically thick accretion disk models predict a strong Lyman edge at 13.6 eV. Yet observed AGN ultraviolet continua are smooth, lacking this feature.

Why is the predicted Lyman edge absent in observed AGN spectra?

What this paper is selling

A 3D radiation MHD simulation of a sub-Eddington disk reveals magnetic pressure support lowers densities, weakening Lyman opacity. This naturally suppresses the edge, reconciling theory with observations.

Key results to remember

- No sharp Lyman edge appears in the simulated spectrum.
- Magnetic pressure support reduces surface density by two orders of magnitude.
- Lower densities weaken the Lyman opacity jump, suppressing the edge.

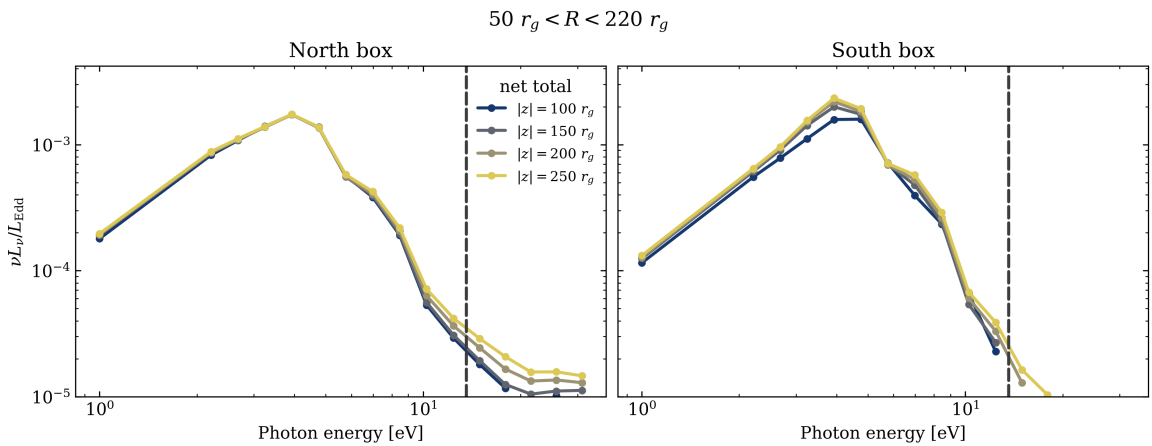


Figure 4 | Simulated spectrum shows no Lyman edge at 13.6 eV, consistent with observations.

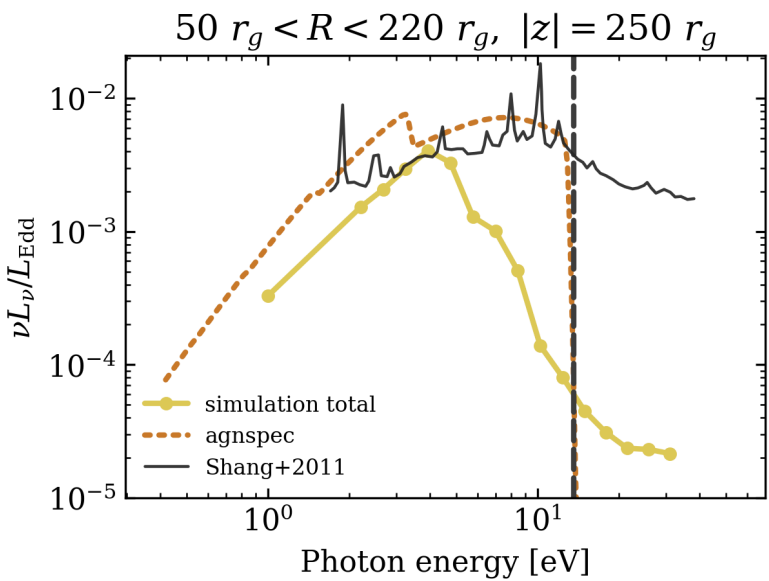


Figure 8 | Simulation spectrum lacks Lyman edge but is underluminous compared to observed quasar composite.